JAVA PRACTICAL PROGRAM REPORT

Total and Average Marks Calculator

Program Name: AverageMarks.java

Language: Java

Topic: Keyboard input, addition, and average calculation

This report describes the program, formulas, algorithm, class diagram, source code, execution steps, and sample console output.

1. Aim

To input marks in three subjects and calculate the total and average marks.

2. Requirements

• Java Development Kit (JDK)
• Text editor or Java IDE
• Save the source file as **AverageMarks.java**

3. Formulas

Total Marks = m1 + m2 + m3

Average Marks = Total Marks / 3.0

Using 3.0 allows the average to include decimal values.

4. Algorithm

1. Start the program.

2. Create a Scanner object.

3. Read marks for the three subjects.

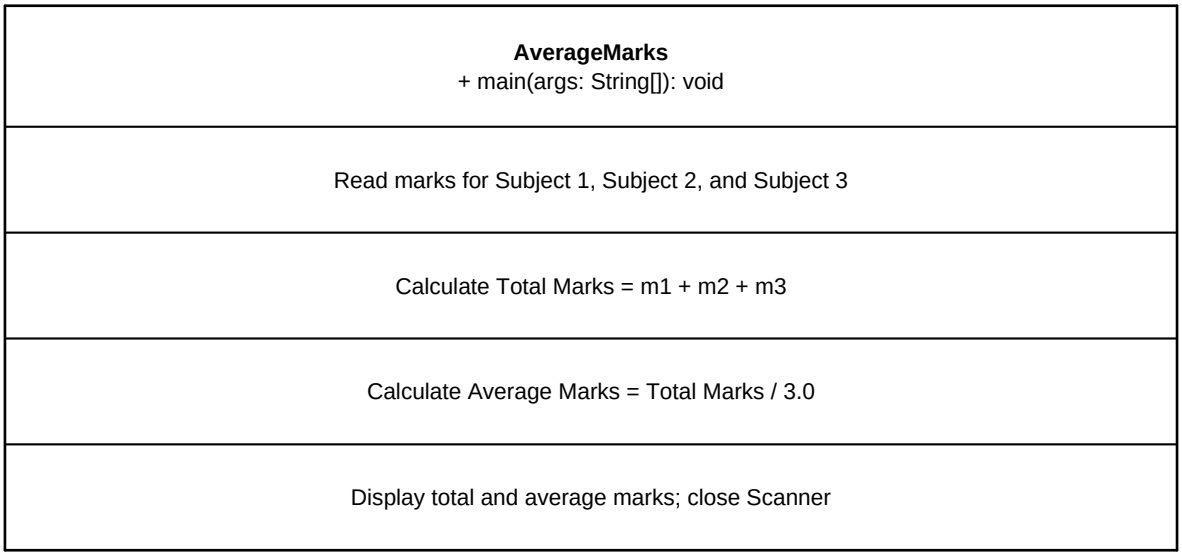
4. Add all three marks to calculate the total.

5. Divide the total by 3.0 to calculate the average.

6. Display the total and average marks.

7. Close Scanner and end.

5. UML / Class Diagram



The class has one main method. It reads three integer inputs, calculates their sum and arithmetic mean, displays both results, and closes the Scanner.

6. Source Code

```
import java.util.Scanner;

public class AverageMarks {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter marks in Subject 1: ");
        int m1 = sc.nextInt();

        System.out.print("Enter marks in Subject 2: ");
        int m2 = sc.nextInt();

        System.out.print("Enter marks in Subject 3: ");
        int m3 = sc.nextInt();

        int total = m1 + m2 + m3;
        double average = total / 3.0;

        System.out.println("Total Marks: " + total);
        System.out.println("Average Marks: " + average);

        sc.close();
    }
}
```

7. Compilation and Execution

```
javac AverageMarks.java
java AverageMarks
```

Sample inputs: Subject 1 = 85, Subject 2 = 90, Subject 3 = 95.

8. Console Output

Captured from a successful compile and run. Inputs are shown beside each prompt for readability.

```
Enter marks in Subject 1: 85
Enter marks in Subject 2: 90
Enter marks in Subject 3: 95
Enter marks in Subject 1: Enter marks in Subject 2: Enter marks in Subject 3: Total Marks: 270
Average Marks: 90.0
```

9. Result / Conclusion

For sample marks of 85, 90, and 95, the total is 270 and the average is 90.0. The UML and console sections have been adjusted so labels and output remain inside their boxes.